

SBC 201 Chapter 5 General Building Heights and Areas — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–16 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 5, source file Reference\SBC 201 2024\source_reference\Chapter_05 – GENERAL BUILDING HEIGHTS AND AREAS.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **1425** independently checkable numeric records (**150** Verified, **1275** Verify source). Unresolved OCR is listed in the register and is not a design-release value.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, height/area calculation package, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapter 4, Chapter 6 / Table 601, Section 403, Section 406 / Table 406.5.4, Section 420, Section 706, Section 707 / Table 707.3.10, Section 716, Section 903, Section 907, Section 1511, SBC 401, SBC 501, SBC 801, SBC 901, NFPA 13/13R/13D manuals, commentary examples, or the existing chapter summary. Where Chapter 5 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications. Chapter 5 does not restate the high-rise definition; Section 403 remains outbound.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Type of construction is unconfirmed. Tables 504.3, 504.4 and 506.2 are concatenated OCR; type-specific metre, story and area cells are **Verify source** except the uniform Group R **S13R / S13D 18 m** strings.
4. Automatic sprinkler protection is not selected. **NFPA 13 / Section 903.3.1.1** (table tags NS / S / S1 / SM) and **NFPA 13R / Section 903.3.1.2** (S13R) are shown as separate branches. Note h to Tables 504.3 / 504.4 / 506.2 states new Group R occupancies shall be sprinklered per 903.2.8; NS cells are for SBC 901 existing-building evaluation.
5. Storey count, grade plane, fire walls, podium 510.2, mixed B/M/S-2/A, occupied roof, mezzanines and live/work units are unconfirmed.
6. Section 507 unlimited-area paths (F/S warehouses, malls, hangars, Group E, A-3 halls) are omitted from lead tables. Live/work (508.5) is omitted unless that program is opened in the gap register.
7. Table 504.4 is page-split after the A-1 start; **no R-2 story cells are present** in the extract.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, construction type, sprinkler branch, mixed-use/podium condition or exception exists.
Not typical	Unrelated occupancy-only or low-rise-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 5 points to another section/code/standard, or OCR/AHJ must be confirmed first.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 5 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 5 application	Required project action
Construction type	Unconfirmed; high-rise R-2 often Type I, not verified here	Tables 504.3 / 504.4 / 506.2 and 510 specials are type-specific; type-to-cell mapping is OCR-unresolved	Freeze type of construction on the code data sheet; read published tables before locking height, stories or area
Sprinkler basis	Unconfirmed: neither NFPA 13 nor 13R is assumed	Selects S vs S13R rows; S13R height in this extract is 18 m for every construction type — below the stated occupied floor > 23 m	Fire engineer to lock 903.3.1.1 vs 903.3.1.2; do not use the S13R height row for a >23 m occupied floor without a published-table check
Storeys and grade plane	Occupied floor stated above 23 m; exact height, grade plane and storey count unconfirmed	Independent tests in 503.1; Table 504.4 R-2 cells are missing; Eq. 5-2 applies above three stories	Issue a signed datum sheet with grade plane, roof height, basements, mezzanines and storey count
Fire walls vs one building	Unconfirmed	503.1 treats each 706 fire-wall portion as a separate building for height, stories, area and type	Show fire walls on the life-safety plan if the massing exceeds one tabular building; do not import 706 ratings here
Podium / 510.2	Unconfirmed mixed parking/retail below	3-hour Type IA plate can split area/stories/type; overall metres still from grade plane; Item 7 unit token is OCR	Freeze whether 510.2 is used; detail the 3-hour assembly, Type IA below, and shaft ratings
Mixed B / M / S-2 / A	Unconfirmed podium and amenity program	508.3 uses the most restrictive height/area; 508.4 uses Table 508.4 hours (flattened) and area ratios ≤ 1	Classify every room; choose accessory / nonseparated / separated / 510 and show it on the code sheet
Occupied roof	Unconfirmed	503.1.4 occupancy vs story-below; enclosure 1200 mm ; penthouses outbound to 1511	State occupied-roof uses and enclosure heights; do not count complying occupied roofs as area or a story
Mezzanines	Unconfirmed	2.1 m clear; aggregate one-third (dwelling-unit one-half branch)	Identify every mezzanine vs story; keep loft units on the 505.2.1 Exception 3 path if used

Decision / gap	Current project basis	Why it controls Chapter 5 application	Required project action
Live/work units	Not opened	508.5 would classify the unit as R-2 with 280 m² / 50% / five workers	Omit 508.5 from lead tables unless a live/work program is added
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	High-rise 403, sprinklers 903 and fire-wall 706 cannot be concluded from Chapter 5 alone	Engage the qualified local/fire consultant before design freeze

4. Scope and independent limits

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
501.1	Chapter height/area scope	This chapter controls the height and area of structures hereafter erected and additions to existing structures	New buildings and additions	None stated	Direct	Evaluate the entire enlarged building against Chapter 5 when an addition is not fire-wall separated	Verified
503.1	Independent height, stories and area	Building height, number of stories and building area shall not exceed Sections 504 and 506 based on Section 602 type and Section 302 occupancy, except as modified in Chapter 4 and this chapter. Apply the three tests independently	Every building and fire-wall portion	Chapter 4 and this chapter modifications, including 507 and 510	Direct	Record type, occupancy, sprinkler tag, height, stories and area as three separate code-sheet lines	Verified
503.1	Fire-wall separate buildings	Each portion separated by one or more fire walls complying with Section 706 is a separate building for area, height, stories and type of construction	Structure divided by fire walls	706 construction is outbound; do not import ratings	Conditional	Stop fire-wall height/area takeoff at each 706 line; verify 706 separately	External verification
503.1.2	Buildings on the same lot	Two or more buildings on the same lot are separate buildings, or may be treated as portions of one building where each building's height and stories and the aggregate area are within Sections 504 and 506	More than one building on the lot	Treating them as one building applies the aggregate-building provisions to each	Conditional	Either assume an imaginary lot line (705.3 outbound) or prove the aggregate still fits 504/506	Verified
503.1.3	Type I unlimited tabular buildings	Type I buildings permitted unlimited tabular height and area are not required to use 507, 503.1.1, 504.3 unlimited-height options, or other-type increases	Type I with unlimited Table 504/506 cells	This is not a release from high-rise 403.3 sprinklers	Conditional	If Type I unlimited cells are confirmed from the published tables, do not add 507 yard conditions as a substitute	External verification

5. Address identification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
502.1	Address character size	Approved address identification, legible and visible from the street or road fronting the property; characters contrast with the background; Arabic numbers or letters, not spelled-out numbers; each character minimum 100 mm high with minimum 12.5 mm stroke width; maintain the identification	New and existing buildings	Fire code official may require additional locations; private-road sites use a monument, pole or other approved means where the address cannot be viewed from the public way	Direct	Place contrasting 100 mm / 12.5 mm address characters on the street elevation and any fire-official extra locations	Verified

6. Occupied roofs

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
503.1.4	Occupied-roof occupancy	A roof level or portion may be an occupied roof if that occupancy is permitted by Table 504.4 for the story immediately below. Occupied-roof area is not building area under 506. An occupied roof is not included in building height or number of stories under 504 where penthouses and other enclosed rooftop structures comply with Section 1511	Occupied roof proposed	Exception 1: occupancy on the roof is not limited to the story below where the building is sprinklered throughout per 903.3.1.1 or 903.3.1.2 and occupant notification per 907.5.2.1 and 907.5.2.3 is provided on the occupied roof; EVACS per 907.5.2.2 is also required on that roof where such a system is required elsewhere in the building. Exception 2: assembly on roofs of Type I or II open parking per the exception to 903.2.1.6	Conditional	Classify the roof use; keep penthouses on 1511; do not add complying occupied-roof area to Table 506.2 takeoff	Verified
503.1.4.1	Occupied-roof enclosure height	Elements or structures enclosing the occupied roof shall not extend more than 1200 mm above the occupied-roof surface	Enclosures around an occupied roof	Exception: penthouses per 1511.2 and towers, domes, spires and cupolas per 1511.5. Mechanical screens and bulkheads are not in that exception	Conditional	Limit guard walls, windscreens and similar enclosures to 1200 mm unless a listed 1511 penthouse/tower path is used	Verified

7. Height in metres

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
504.1	Height attributes	Height in metres and number of stories are determined from type of construction, occupancy and whether an automatic sprinkler system is installed throughout	Every building	Exception: one-story aircraft hangars / paint hangars / aircraft manufacturing with Chapter 9 suppression and yards/public ways not less than one and one-half times the building height — not a conventional R-2 path	Direct	Lock occupancy, type and sprinkler tag before reading Table 504.3	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
504.2	Mixed-occupancy height	In a mixed-occupancy building per 508, no individual occupancy shall exceed this section's height and story limits for that occupancy	Mixed occupancy	Actual building height depends on 508.3 vs 508.4	Conditional	Check each occupancy against 504; do not let a more restrictive podium occupancy sit above its own story cap	Verified
504.3	Tabular height in metres	Maximum building height in metres shall not exceed Table 504.3	All buildings	Rooftop exception below. Table cells are concatenated; construction-type headers are garbled	Direct	Use Table 504.3 only after the published grid is verified; do not reconstruct type columns from memory	Verify source
504.3 Exception	Combustible rooftop structures	Towers, spires, steeples and other rooftop structures shall match the building's required type except where 1511.2.4 permits otherwise; not for habitation or storage. Noncombustible such structures are unlimited in height; combustible such structures shall not extend more than 6 m above the allowable building height	Rooftop towers/spires/steeples	Habitation or storage makes the structure a story	Conditional	Keep combustible architectural rooftop elements within 6 m of the Table 504.3 cap; send penthouse geometry to 1511	Verified
Table 504.3	Group R S13R/S13D height	Group R S13R and S13D rows in the extract are twelve consecutive 18 values, so 18 m above grade plane for every listed construction type on those sprinkler tags	Group R with 903.3.1.2 (S13R) or 903.3.1.3 (S13D) throughout	S13D is not the typical R-2 high-rise system. NS and S (903.3.1.1) type-specific metre cells remain unmapped	Direct	Do not place an occupied floor above 23 m on the S13R 18 m row. Confirm S (NFPA 13) type-specific metres from the published table	Verified
Table 504.3 note h	New Group R sprinklers	New Group R occupancies shall be protected by an automatic sprinkler system in accordance with Section 903.2.8	New Group R	NS values are only for existing-building height evaluation per SBC 901 (note d)	Direct	Design new R-2 on an S or S13R row, not NS; verify 903.2.8 separately	Verified

8. Number of stories

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
504.4	Tabular stories	Maximum number of stories above grade plane shall not exceed Table 504.4	All buildings	Table starts at A-1 then page-splits; no R-2 story cells are in the extract	Direct	Hold storey count until the published Table 504.4 Group R row is read; do not invent R-2 story limits	Verify source
Table 504.4 note h	New Group R sprinklers	New Group R occupancies shall be protected by an automatic sprinkler system in accordance with Section 903.2.8	New Group R	NS for existing buildings per SBC 901 (note d)	Direct	Same sprinkler lock as Table 504.3 note h; do not reuse an NS story cell for new R-2	Verified

9. Mezzanines and equipment platforms

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
505.2	Mezzanine story/area treatment	A complying mezzanine is a portion of the story below; it does not contribute to building area or number of stories under 503.1. Include mezzanine area in the fire area. Clear height above and below the mezzanine floor not less than 2.1 m	Mezzanine proposed	Means of egress follows Chapter 10 (outbound)	Conditional	Keep 2.1 m clear both sides; do not add a complying mezzanine to Table 506.2 area or Table 504.4 stories	Verified
505.2.1	Mezzanine aggregate area	Aggregate mezzanine area within a room not greater than one-third of that room's floor area. Enclosed portions of the room are excluded from the room area. Mezzanine area is not included in the room area for this check	Mezzanine in a room	Exception 1: Type I/II special industrial 503.1.1 — two-thirds (not typical). Exception 2: Type I or II throughout with 903.3.1.1 and 907.5.2.2 EVACS — one-half . Exception 3: mezzanine in a dwelling unit in a 903.3.1.1 or 903.3.1.2 building — one-half , if open except bathrooms/closets, opening unobstructed except walls not more than 1050 mm , columns and posts, and 505.2.3 exceptions are not used	Conditional	Use one-third for amenity mezzanines unless Exception 2 is proven; use the 1050 mm dwelling-unit path only for loft units	Verified
505.2.3	Mezzanine openness	Mezzanine shall be open and unobstructed to the room except for walls not more than 1 m in height, columns and posts	Mezzanine	Exception 1: enclosed occupant load not greater than 10 . Exception 2: two or more exits or access to exits. Exception 3: enclosed area not greater than 10 percent of the mezzanine. Exception 4: industrial control glazing. Exception 5: other than Groups H and I, no more than two stories , 903.3.1.1 throughout, two or more exits — not a high-rise path	Conditional	Keep common-area mezzanines open above 1 m unless a numbered exception is documented	Verified
505.2.1.1–505.3.1	Mezzanine plus equipment platform	Where a room contains both, aggregate area of mezzanines and equipment platforms not greater than two-thirds of the room; mezzanine still limited by 505.2.1. Equipment platforms alone not greater than two-thirds of the room. Platforms are not a story or building area and are not in the 903 fire area	Equipment platform, or both in one room	Platform access shall not serve building egress. Guards per 1015.2 (outbound)	Conditional	Limit plant platforms to two-thirds of the plant room; sprinkle above and below where 903.3 requires (505.3.2)	Verified

10. Allowable area method

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
506.1.3	Basement area exclusion	Basements need not be included in the total allowable floor area provided the total basement area does not exceed the area permitted for a one-story-above-grade-plane building	Basements present	Multiple basements are aggregated against that one-story allowable	Conditional	Compare aggregate basement area to the one-story allowable (including 506.3 frontage); count basements in the building total if they exceed it	Verified
506.1.3.1	Full-site basement	One or more basement floors covering the entire land area are permissible provided public-safety, ventilation, groundwater drainage and sewage requirements of the relevant executive regulations and environmental requirements are met	Basement proposed under the full site	Technical/architectural/structural requirements are not quantified here	Conditional	Coordinate full-site basement with those executive/environmental rules; do not treat this clause as extra Table 506.2 area	Verified
506.2.1	Single-occupancy area equations	Allowable area of each story: $(Aa=[At+(NS \times lf)])$ (Equation 5-1). <i>Total allowable area of a building more than three stories above grade plane: that quantity times (Sa) (Equation 5-2), where (Sa=3), or (Sa=4) where the building is equipped throughout per 903.3.1.2. No individual story shall exceed Equation 5-1</i>	Single-occupancy building	<i>(At) and NS come from Table 506.2 (OCR). Frontage (lf) from 506.3</i>	Direct	Run 5-1 per story and 5-2 for the tower total; do not use (S_a=4) unless 13R is actually installed	Verified
506.2.2	Mixed-occupancy area ratios	Each story follows 508.3.2 or 508.4.2 as applicable. For buildings more than three stories , the aggregate sum of (actual story area / allowable story area) using Equation 5-3 shall not exceed three	Mixed occupancy, more than three stories	Exception: separated occupancies per 508.4 and sprinklers throughout per 903.3.1.2 — the ratio sum shall not exceed four	Conditional	Sum story ratios and keep the total ≤ 3 (or ≤ 4 only on a proven 13R separated mixed building)	Verified
Table 506.2	Group R-2 tabular (A_t)	Table 506.2 supplies NS, S1, SM, S13R and S13D area factors in square metres. The R-2h block is concatenated; construction-type assignment is unresolved	R-2 area takeoff	Note h: new Group R sprinklers per 903.2.8. Note d: NS for SBC 901 existing buildings. Note i (Group U greenhouse 435 m²) is not an R-2 factor	Direct	Do not adopt R-2 (A_t) cells from the flattened grid; verify the published R-2 row before area calculations	Verify source

11. Frontage increase

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
506.3.1	Frontage perimeter threshold	To qualify for a frontage area-factor increase, not less than 25 percent of the building perimeter shall be on a public way or open space. The open space shall be on the same lot or dedicated for public use and accessed from a street or approved fire lane	Frontage increase claimed	Fire-wall edges are perimeter, not open frontage	Direct	Measure open perimeter including fire-lane access; do not count inaccessible yards	Verified
506.3.2	Minimum open-space width	Public way or open space used for the increase shall have a minimum distance of 6 m measured at right angles from the building face to the closest interior lot line, the entire width of a street/alley/public way, or the exterior face of an adjacent building on the same property. The increase uses the smallest qualifying width that is 6 m or greater and the percentage of perimeter that has at least 6 m	Frontage increase claimed	Width is the full space between buildings on the same lot, not fire-separation distance to an imaginary line	Direct	Dimension perpendicular yards/streets on the site plan; take (L_f) from the smallest ≥ 6 m side	Verified
Table 506.3.3	Frontage increase factor	(L_f) is taken from Table 506.3.3. Interpolation within the table is permitted	Building qualifies under 506.3.1–506.3.2	Table is flattened (percent $\times$ open-space columns concatenated)	Direct	Verify published 506.3.3 factors before applying (L_f) in Equations 5-1 / 5-2 / 5-3	Verify source
Table 506.3.3.1	Near-unlimited-area frontage	Where a building meets Section 507 except the 18 m public-way/yard width, (L_f) is taken from Table 506.3.3.1	Failed 507 yard width only	Not a typical R-2 high-rise path. Table is flattened	Conditional	Do not use 506.3.3.1 unless a 507 configuration is actually proposed	Verify source

12. Mixed occupancy

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
508.1	Mixed-use options	Each portion is classified per 302.1. Where more than one occupancy group exists, comply with 508.2, 508.3, 508.4, 508.5 or a combination	More than one occupancy group	Exception 1: occupancies separated per 510. Exception 2: Table 415.6.5 detached Group H-1/H-2/H-3	Conditional	State accessory / nonseparated / separated / 510 on the code sheet; do not mix methods silently	Verified
508.2.3	Accessory occupancy area	Building allowable area is per 506 for the main occupancy. Aggregate accessory occupancies shall not occupy more than 10 percent of the floor area of the story and shall not exceed Table 506.2 nonsprinklered tabular values for each accessory occupancy	Accessory occupancies ancillary to the main use	Height/stories follow 504 for the main occupancy (508.2.2). No separation required except 508.2.4 exceptions	Conditional	Keep amenity/accessory fire areas $\leq 10\%$ of the story and $\leq$ the NS tabular cap for that accessory group	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
508.2.4 Ex. 2	Dwelling-unit accessory separation	Group I-1, R-1, R-2 and R-3 dwelling units and sleeping units shall be separated from other dwelling/sleeping units and from accessory occupancies contiguous to them per Section 420	Accessory space contiguous to an R-2 unit	Exception 1: Group H-2 through H-5 still need 508.4 separation	Direct	Do not waive 420 unit separations because a lounge or office is "accessory"	External verification
508.3.2	Nonseparated height and area	Allowable building area, height and stories of the nonseparated portion are the most restrictive allowances of the occupancy groups under consideration for the building's type, per 503.1	Nonseparated mixed occupancy	Most restrictive Chapter 9 applies throughout the nonseparated area (508.3.1). High-rise: most restrictive 403 of those occupancies applies throughout (508.3.1.1)	Conditional	Size the nonseparated volume to the worst occupancy's 504/506 row; apply 403/Chapter 9 on that same volume	Verified
508.3.3 Ex. 2	Nonseparated dwelling-unit separation	R-2 dwelling/sleeping units still separate from other units and contiguous occupancies per Section 420	Nonseparated mixed building with R-2 units	Group H-2 through H-5 still need 508.4	Direct	Keep 420 unit and occupancy separations even on the nonseparated path	External verification
508.4.2	Separated occupancy area ratios	In each story, the sum of (actual area of each separated occupancy / allowable area of that occupancy) shall not exceed 1	Separated mixed occupancy	Frontage increase uses the entire building perimeter	Conditional	Tabulate occupancy areas and allowables per story; keep the ratio sum ≤ 1	Verified
508.4.3	Separated occupancy height	Each separated occupancy shall comply with 503.1 height and story limits for the building's type of construction	Separated mixed occupancy	Exception: Section 510 specials	Conditional	Park each occupancy only on stories Table 504.4 allows for that group	External verification
Table 508.4	Occupancy separation hours	Required fire-barrier / horizontal-assembly hours between occupancies, S vs NS columns. S means 903.3.1.1 throughout; NS includes buildings with 13R or 13D. N = no separation; NP = not permitted	Separated occupancies	Note a: Section 420. Note b: private/pleasure-vehicle areas reduced 1 hour but not less than 1 hour . Note f: fire-area separations also meet 707.3.10 / Table 707.3.10. Table is flattened	Conditional	Verify published R vs B/M/S-2/A hours before rating podium separations; do not adopt concatenated 1/2/3 hour cells	Verify source
508.4.4.1	Mass-timber occupancy thermal barrier	Required occupancy separations shall be 707 fire barriers or 711 horizontal assemblies, or both. In Type IV-B or IV-C, mass-timber elements serving those separations shall have an approved thermal barrier of gypsum board not less than 12.5 mm thick, or a material meeting both NFPA 275 Temperature Transmission and Integrity tests, toward the interior	Type IV-B/IV-C occupancy separations	Not applicable unless mass timber is the construction type	Conditional	Specify 12.5 mm gypsum or NFPA 275 thermal barrier on the interior face of mass-timber occupancy separations	Verified

13. Incidental uses

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
509.1	Incidental-use scope	Incidental uses in Table 509.1 shall comply with this section. They are ancillary functions limited to that table	Listed incidental room or system	Exception: incidental uses within and serving a dwelling unit need not comply	Direct	Apply Table 509.1 to common-building plant/laundry/waste; not to in-unit laundry	Verified
509.2	Incidental occupancy class	Incidental uses are not individually classified per 302.1; they take the building occupancy in which they are located	Incidental use present	They do not create a mixed occupancy by themselves	Direct	Keep laundry/waste/plant rooms as R-2 for occupancy classification	Verified
509.3	Incidental area cap	Incidental uses shall not occupy more than 10 percent of the building area of the story in which they are located	Incidental uses on a story	None stated	Direct	Limit each story's Table 509.1 rooms to 10% of that story	Verified
Table 509.1	Furnace and boiler rooms	Furnace room where any piece of equipment is over 117 kW per hour input: 1 hour or an automatic sprinkler system. Rooms with any boiler over 103.5 kPa and 7.5 kW: 1 hour or an automatic sprinkler system	Those equipment rooms outside dwelling units	Sprinkler need only protect the incidental room (509.4.2.1 / note a)	Conditional	Rate or sprinkle furnace/boiler rooms at those thresholds; show smoke-resisting enclosure if the sprinkler-only path is used (509.4.2)	Verified
Table 509.1	Laundry and waste rooms	Laundry rooms over 9 m²: 1 hour or an automatic sprinkler system. Waste and linen collection rooms over 9 m² (other than ambulatory care and Group I-2): 1 hour or an automatic sprinkler system	Common laundry or waste rooms over 9 m²	I-2 / ambulatory rows are a different (stricter) branch. Dwelling-unit exception under 509.1	Direct	Provide 1-hour enclosure or room sprinklers for common laundry and waste rooms larger than 9 m²	Verified
Table 509.1	Hydrogen rooms in Group R	Hydrogen fuel gas rooms not classified as Group H: 1 hour in Groups B, F, H, M, S and U; 2 hours in Groups A, E, I and R	Hydrogen fuel gas room in an R occupancy	Not classified as Group H	Conditional	If a hydrogen room is provided in the R-2 building, use the 2-hour Group R rating	Verified
509.4.1.1	Mass-timber incidental thermal barrier	Where Table 509.1 requires a rated separation, mass-timber fire barriers or horizontal assemblies in Type IV-B or IV-C shall have gypsum not less than 12.5 mm or NFPA 275 thermal barrier toward the interior of the incidental use	Type IV-B/IV-C incidental separations	None stated	Conditional	Same thermal-barrier detail as 508.4.4.1, on the incidental-use face	Verified

14. Podium and parking specials

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
510.2	Horizontal building separation	Upper and lower volumes are separate buildings for area, fire-wall continuity, stories and type where all listed conditions are met: 3-hour horizontal assembly (vertical offsets and their supports also 3 hours); building below, including the assembly, is Type IA; shafts/stair/ramp/escalator enclosures through the assembly not less than 2 hours with 716 protectives; occupancies above limited to Group A uses each with occupant load less than 300 , or Groups B, M, R or S; building below sprinklered throughout per 903.3.1.1 and not Group H	510.2 podium/pedestal elected	Item 3 exception: 3-hour enclosure below and 1-hour above where the building above is not required to be Type I, the enclosure connects fewer than four stories , and opening protectives above are not less than 1 hour . Item 4: combustible interior exit stairs in the Type IA building only if the building above is Type III, IV or V and the stair in the Type IA building is in 3-hour construction	Conditional	Detail the 3-hour Type IA plate, 2-hour shafts, occupancy list and NFPA 13 below; count upper stories from the plate	Verified
510.2 Item 7	Podium overall height unit	Maximum building height in mm shall not exceed Section 504.3 for the building having the smaller allowable height, measured from the grade plane	510.2 podium	The printed unit is mm in the extract; do not treat it as millimetres and do not silently convert	Conditional	Verify the published unit for Item 7 before locking overall podium height from grade plane	Verify source
510.4	Parking beneath Group R	Where a maximum one story above grade plane Group S-2 parking garage (enclosed or open, or combination), of Type I construction or open of Type IV, with grade entrance, is provided under Group R, the number of stories used to determine minimum type is measured from the floor above that parking. The floor assembly shall comply with the parking garage's type and with Table 508.4 mixed-occupancy separation	One-story S-2 parking under R	Table 504.3 height in metres is not increased. Table 508.4 hours are OCR	Conditional	If this path is used, start the R story count above the garage; keep the metre cap from Table 504.3	Verified
510.9	Multiple buildings on a pedestal	Where two or more buildings sit above the horizontal assembly under 510.2, 510.3 or 510.8, those upper buildings are separate from each other and comply as distinct buildings	Multiple towers or fire-wall-split buildings on one plate	None stated	Conditional	Treat each upper building's type, height and area independently on the 510.2 plate	Verified

15. Alternate R-2 height paths

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
510.5	Type IIIA R-2 height increase	For Type IIIA Groups R-1 and R-2, Table 504.3 height shall be increased by 3 m and Table 504.4 stories increased by one where the first-floor assembly above the basement is not less than 3 hours and floor area is subdivided by 2-hour fire walls into areas of not more than 279 m²	Type IIIA R-2 using this increase	Base Table 504.3/504.4 cells remain OCR. High-rise 403 is still outbound	Conditional	Use only on Type IIIA; show 3-hour first floor, 2-hour fire walls and 279 m² compartments	Verified
510.6	Type IIA R-2 height increase	Height limitation for Type IIA Groups R-1 and R-2 shall be increased to nine stories and 30 m where the building is separated by not less than 15 m from any other building on the lot and from lot lines, exits are segregated in an area enclosed by a 2-hour fire wall, and the first-floor assembly is not less than 1½ hours	Type IIA R-2 using this increase	A building at 30 m may be high-rise; Section 403 is outbound and not waived here	Conditional	Use only on Type IIA with 15 m yards, 2-hour exit fire walls and 1½-hour first floor; still run 403 if high-rise	Verified

16. Open-parking mixed

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
510.7	Open parking beneath Group R	Open parking garages under Groups A, I, B, M and R shall not exceed height and area permitted under Section 406.5. The portion above shall not exceed Section 503 for the upper occupancy. Height in metres and stories of the upper portion is measured from grade plane and includes the open parking garage	Open S-2 parking under R	Table 406.5.4 values are not imported	Conditional	Measure R height and stories from grade through the garage; verify 406.5 separately	External verification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
510.7.1	Open-parking / R separation and egress	Fire barriers or horizontal assemblies between parking and the upper occupancy shall match Table 508.4. Type applies to each occupancy individually, except structural members including main bracing in the garage that support the upper occupancy take the more restrictive Table 601 rating. Means of egress for the upper occupancy shall be separated from parking by fire barriers or horizontal assemblies having not less than a 2-hour fire-resistance rating, with self-closing 716 doors	510.7 configuration	Table 508.4 and Table 601 are outbound/OCR. Parking egress follows 406.5	Conditional	Enclose R exits through the garage in 2-hour construction; do not import 601 or 508.4 hours from this extract	Verified
510.8	B or M under open parking	Group B or M below a lesser-type Group S-2 open parking garage are separate buildings for type where listed conditions are met, including a not less than 2-hour horizontal assembly, Type IA below (exception: Type IB or II, not less than the garage type, where the building below is not greater than one story above grade plane), and garage exits discharging at grade in 2-hour fire barriers or horizontal assemblies	Open parking above B/M podium	Not an R-above path. 406.5 still governs the garage including the height of the building below	Conditional	Use only if retail/office sits under open parking; keep R residential on 510.2 / 510.4 / 510.7 instead	Verified

17. Project-use controls

1. Use **Verified** rows for schematic height/area scoping after the row trigger and sprinkler/type branch are confirmed.
2. Treat every **Verify source** row (flattened Tables 504.3 type-specific cells, all of Table 504.4 including missing R-2 stories, Table 506.2 (A_t), Tables 506.3.3 / 506.3.3.1, Table 508.4 hours, 510.2 Item 7 **mm** token) as a design hold point. No affected value is to be placed in issued-for-approval documents without a published-source check.
3. Do not reconstruct Table 504.3 / 504.4 / 506.2 construction-type columns from IBC memory. The canonical failure is adopting a type-specific metre or area from a concatenated string.
4. Do not assume NFPA 13 versus 13R. The extract’s Group R S13R height is **18 m** for all types — incompatible with a stated occupied floor > **23 m** unless the published table is shown to differ.
5. Do not import Section 403 high-rise, Section 420 unit separation, Table 601, Section 706 fire-wall, Table 406.5.4, 903, 907 or 1511 values.
6. Do not use Section 507 unlimited-area yards, hangar 1.5× height, or 508.5 live/work limits unless those programs are opened in the gap register.
7. Record construction type, sprinkler standard, grade plane, podium election and mixed-use method in the project Golden Thread. This matrix is not evidence of SCD NOC, SBPS approval, or stamped compliance.

18. Coverage summary

Internal inventory of the attached Chapter 5 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 1425
- **Verified:** 150
- **Verify source:** 1275

Counts by top-level section

Top-level section	Records
501	0
502	2
503	1
504	218
505	13
506	914
507	32
508	188
509	28
510	29

Appended-table coverage

Appended table	Records	Verify source records
Table 504.3	192	168
Table 504.4	24	24
Table 506.2	862	861

Appended table	Records	Verify source records
Table 506.3.3	16	16
Table 506.3.3.1	24	24
Table 508.4	180	180
Table 509.1	26	0

Coverage cross-check against SBC 201 Chapter 5 General Building Heights and Areas (2024)_CS.md was topics-only: independent H/S/A tests; occupied roofs; sprinkler-row tags; mezzanines; frontage; mixed occupancy; incidental uses; podium 510.2. No CS.md value was copied into a matrix cell. Commentary figures and worked examples (Type VB office metres/stories, motel area, accessory 1000 m² sketches) were not inventoried.

Table 504.4 remainder after the A-1 start, including every Group R story cell, is absent from the extract and is not counted as invented cells. Table 506.2 page 1 occupancy blocks after A-1 are OCR-cloned; I-1 / I-2 / I-3 / H-5 area rows are not in the continuation table.

19. Unresolved-source register

Hold points for the **1275 Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Table 504.3 type-specific cells (168)	One collapsed HTML row; headers garbled (TYPE IVTYPE VABABABABCHTAB); NS and S (903.3.1.1) metre values cannot be assigned to IA–VB without reconstruction	Adopt only the uniform Group R S13R/S13D 18 m strings. Verify published type columns before locking tower height
Table 504.4 (24 visible + remainder missing)	Table starts at A-1 NS/S then page-splits to footnotes; no R-2 story cells	Do not invent R-2 story limits. Read the published table before freeze
Table 506.2 (861)	Two concatenated pages; page 1 occupancy blocks after A-1 are cloned; R-2h block exists but columns unmapped; I-1/I-2/I-3/H-5 rows missing	Do not use flattened (A_t) in Equations 5-1 / 5-2 / 5-3
Tables 506.3.3 and 506.3.3.1 (40)	Percent × open-space factor grids concatenated; 506.3.3.1 title OCR-labelled “SECTION 507 BUILDINGS”	Keep 25% / 6 m from 506.3 prose; verify published (I,f) cells
Table 508.4 (180)	S/NS hour matrix flattened (122NP, 34232 digit runs)	Verify published R vs B/M/S-2/A hours; note b 1-hour floor remains usable as text
510.2 Item 7	Prints height in mm	Do not treat as millimetres and do not convert to metres from memory
507.10 yard width (1)	“18 m in width than one and one-half times the building height” is garbled	Unlimited-area hangar path omitted from lead tables; not an R-2 check